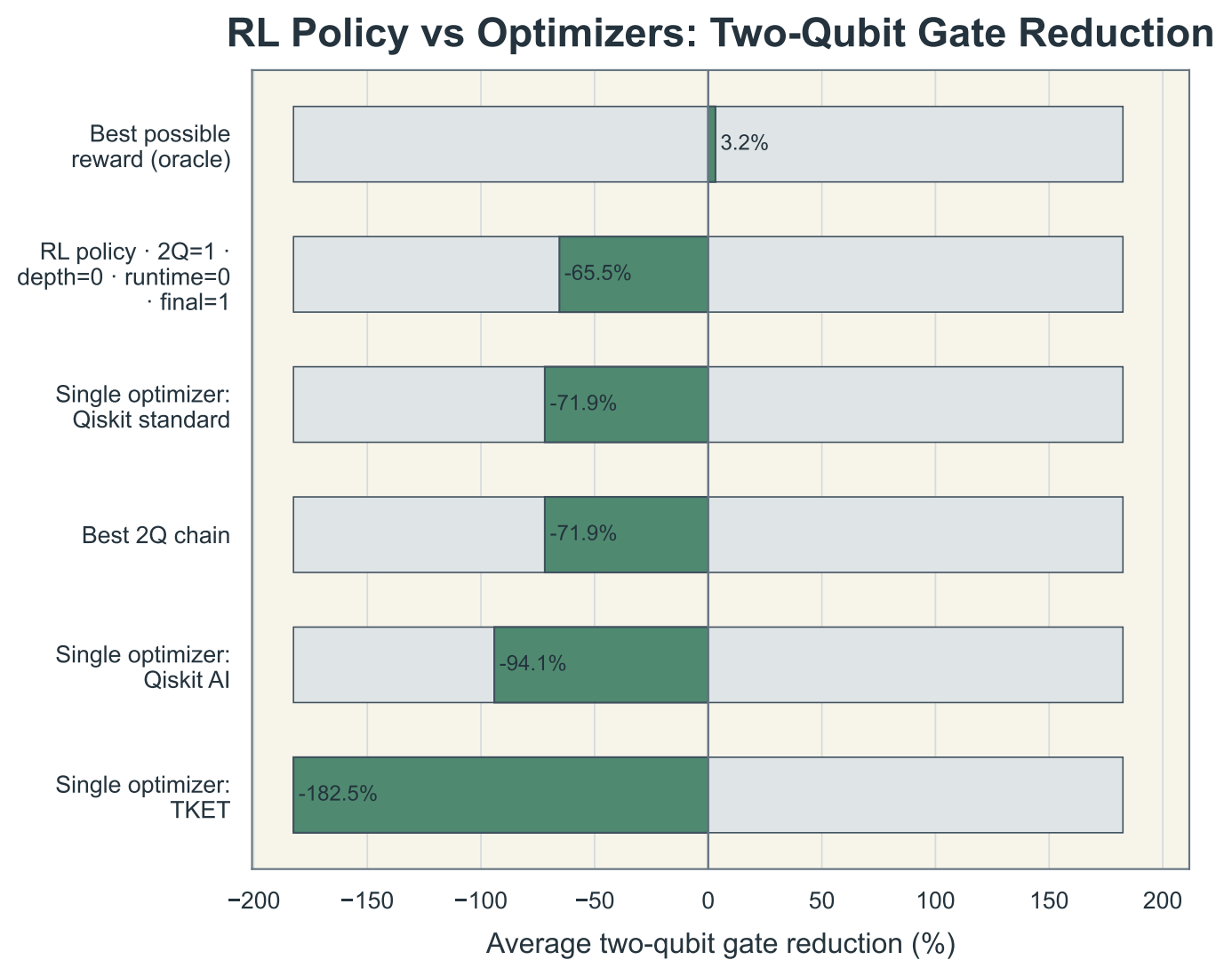
RL Policy vs Optimizers: Two-Qubit Gate Reduction
Best possible reward (oracle)
RL policy · 2Q=1 · depth=0 · runtime=0 · final=1
Single optimizer: Qiskit standard
Best 2Q chain
Single optimizer: Qiskit AI
Single optimizer: TKET
3.2%
-65.5%
-71.9%
-71.9%
-94.1%
-182.5%
−200
−150
−100
−50
0
50
100
150
200
Average two-qubit gate reduction (%)